Media Mix Modelling

Measuring marketing effectiveness for a fashion retail client

DATA

117 weeks of sales, marketing & macro data

METHOD

OLS regression with adstock & log transforms

OBJECTIVE

Identify channel ROI & optimise budget allocation

Data & Processing

DATA

Weekly file: 117 rows - sales, marketing spend, competition, weather, brand awareness

Monthly file: 36 rows - unemployment, GDP, consumer confidence, tourists

Merged via year-month key. Monthly values repeat for each week within the month.

TRANSFORMATIONS

Log transforms applied to sales, marketing spend, economic & competition variables

Why: comparable scales + coefficients become elasticities (% change interpretation)

Marketing channels: log(x+1) to handle zero-spend weeks

ADSTOCK

TV (decay 0.5): longer carryover. Correlation improved 0.05 → 0.20

Press (decay 0.3): shorter-lived effect, print fades faster

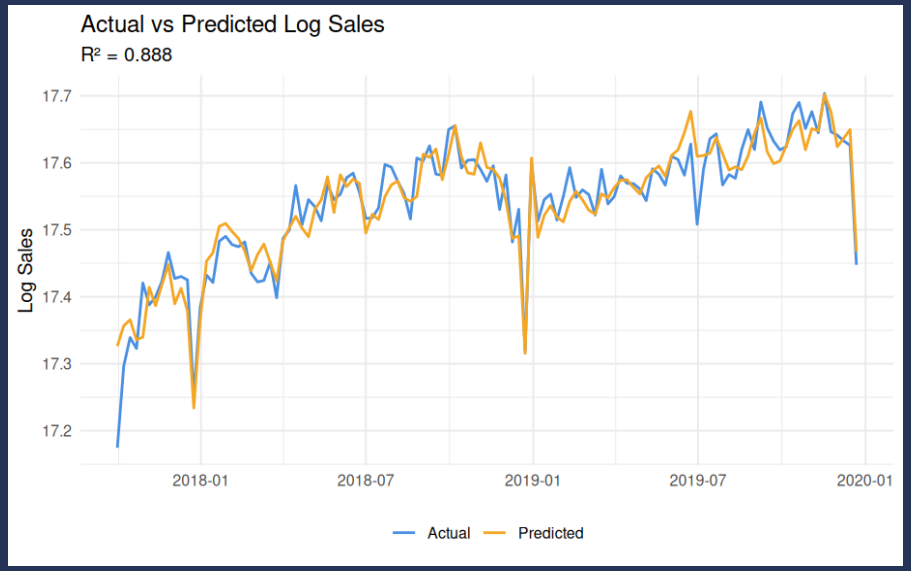
Not applied to: online (direct response), banners (immediate), radio (insufficient data)

Not transformed: Public holidays & Christmas dummy (binary), brand knowledge & competitor recognition (proportions 0–1), weather & precipitation indices.

Final Model Structure

VARIABLE	GROUP	COEFFICIENT	P-VALUE	VIF	SIGN
Online	Marketing	+0.013	0.013 *	1.42	✓ Expected
Banners	Marketing	0.000	0.559	1.6	
TV	Marketing	+0.000	0.612	2.03	
Press (adstock)	Marketing	-0.004	0.013 *	1.50	⚠ Reverse causality
Economy Index	Economic	+0.795	<0.001 ***	2.56	✓ Expected
Unemployment	Economic	-0.653	<0.001 ***	4.82	✓ Expected
Consumer Confidence	Economic	+0.897	<0.001 ***	2.65	✓ Expected
Tourists	Economic	-0.134	<0.001 ***	3.04	✓ Locals crowded out
Competitor Recognition	Competition	-0.210	0.005 **	1.26	✓ Expected
Weather Index	External	-0.001	0.001 **	2.38	✓ Expected
Public Holidays	External	-0.021	<0.005 **	2.14	✓ Expected
Christmas Dummy	External	-0.103	<0.001 ***	2.16	✓ Expected

Model Validation



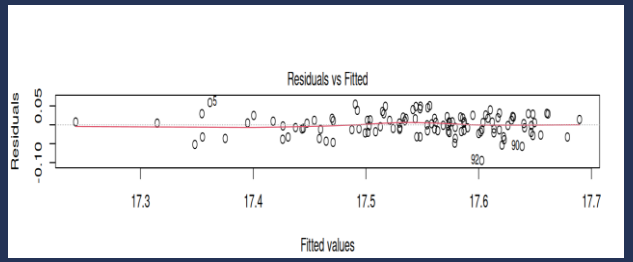
Model tracks actual sales closely across the full two-year period

DIAGNOSTIC CHECKS

Adjusted R^2
0.888
✓ **88.8% of sales variation explained**

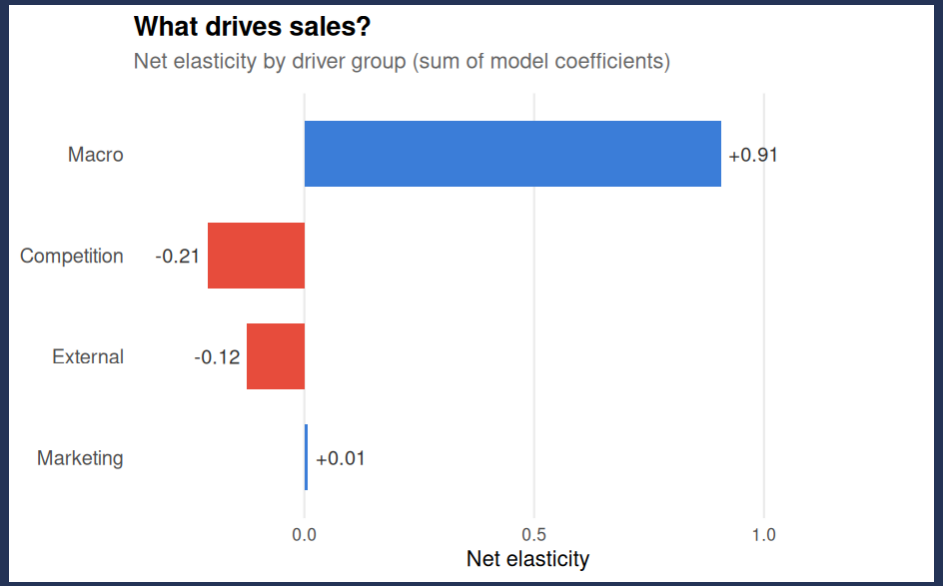
Durbin-Watson
1.33
Mild autocorrelation

VIF
All < 10
✓ **No multicollinearity issues**



What Drives Sales?

Sales in this category are macro-driven, with marketing as a smaller but controllable lever.



Macro is the dominant driver - Economic conditions, consumer confidence, and unemployment together explain most of the variation in sales. When the economy moves, your sales move with it.

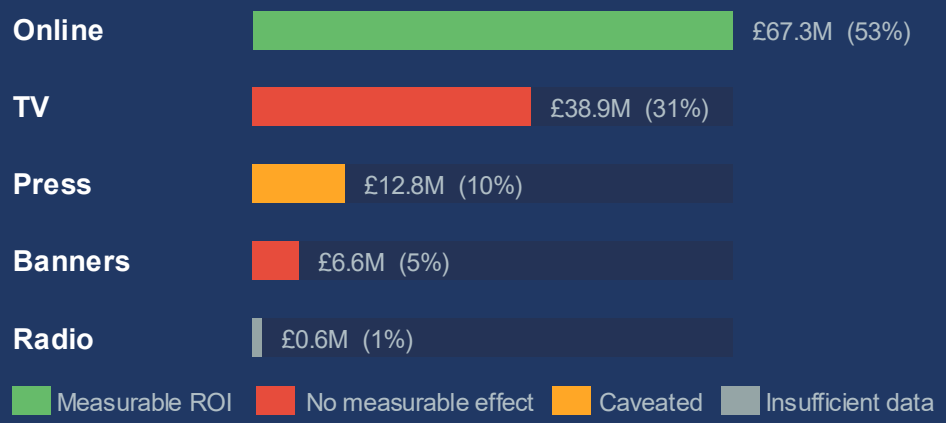
Competitor activity is a drag - Increases in competitor brand recognition measurably reduce sales.

Holidays and weather create volatility - Sales dip around public holidays and Christmas.

Marketing is small in aggregate - Net marketing effect is modest because some channels work and others don't.

Marketing Effectiveness & Recommendations

BUDGET ALLOCATION



CHANNEL VERDICT

Online
Maintain or increase. Only channel with positive measurable impact

TV
No measurable impact despite 31% of budget. Question allocation, but may have long-term brand effects

Press
Negative coefficient reflects reactive spending (ramps up when sales dip), not true negative ROI

Banners
No measurable effect on sales

Radio
Only active 6/117 weeks - insufficient data to assess. Needs a sustained campaign to measure effect